

Falcon-CLIP: Sharp, Smart, and Robust Anomaly Detection



NTHU CVPDL Final Project – Group9

109003814 林士智

114064545 謝尚哲

112064520 陳若瑛

114033636 蔡旻諾

113034506 李家欣

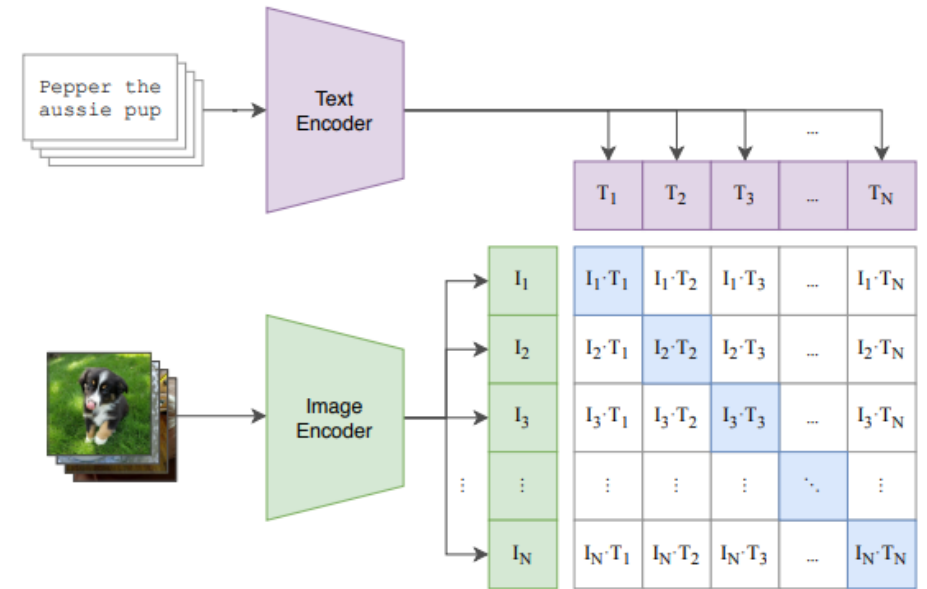
Outline

- Introduction
- Motivation
- Related Work
- Methodology
- Experiment Setup
- Experimental results
- Conclusion

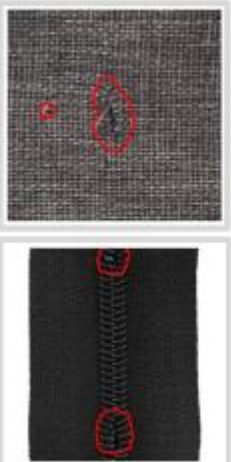
1. Introduction – CLIP [1]

- CLIP **jointly trains an image encoder and a text encoder** to predict the correct pairings of a batch of (image, text) training examples (Fig.1)
- Enables strong zero-shot recognition via image–text similarity
- **Limitation: Anomaly-Unawareness**

Despite the obvious anomalies present in the images (Fig.2), image features have higher similarities to normal descriptions, rather than anomaly descriptions [2]



▲ Fig.1 Image–Text Cosine Similarity Matrix



“This is a [] carpet.”

Semantics	Similarity	Probability _{$\tau=0.01$}
broken	0.18	0.22
normal	0.19	0.78

“This is a [] zipper.”

Semantics	Similarity	Probability _{$\tau=0.01$}
broken	0.20	0.38
normal	0.21	0.62

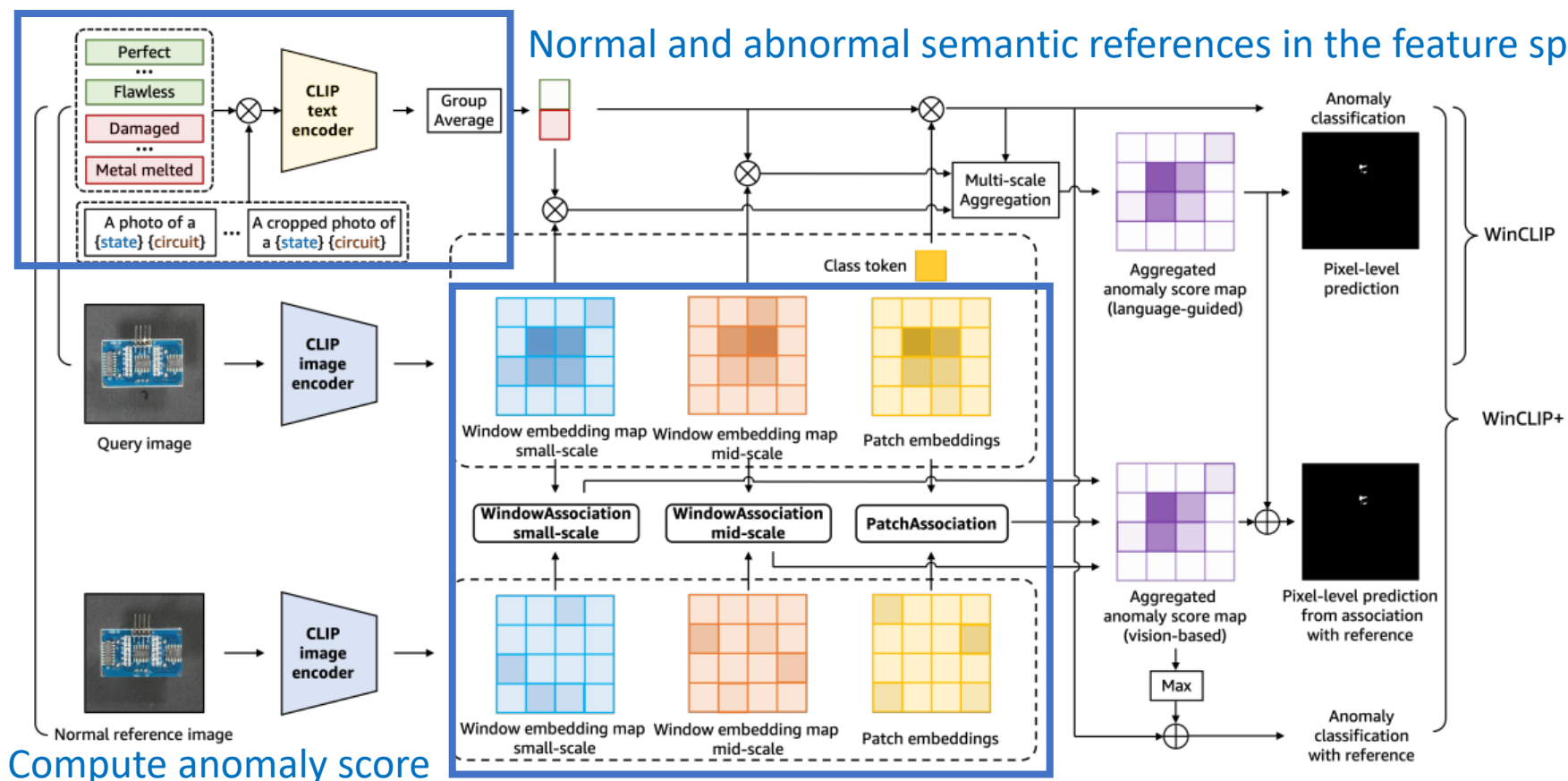
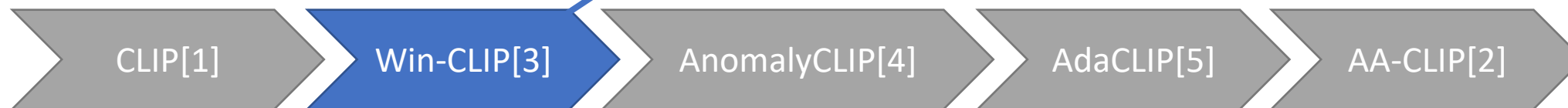
▲ Fig.2 Examples of CLIP Anomaly-Unawareness

2. Motivation

- CLIP lacks explicit anomaly awareness
 - Anomalous regions may be misclassified as normal
 - This limitation is critical in real-world anomaly detection, where abnormal samples are rare and diverse
- We want to build an anomaly-aware model to **better distinguish normal and anomalous regions**

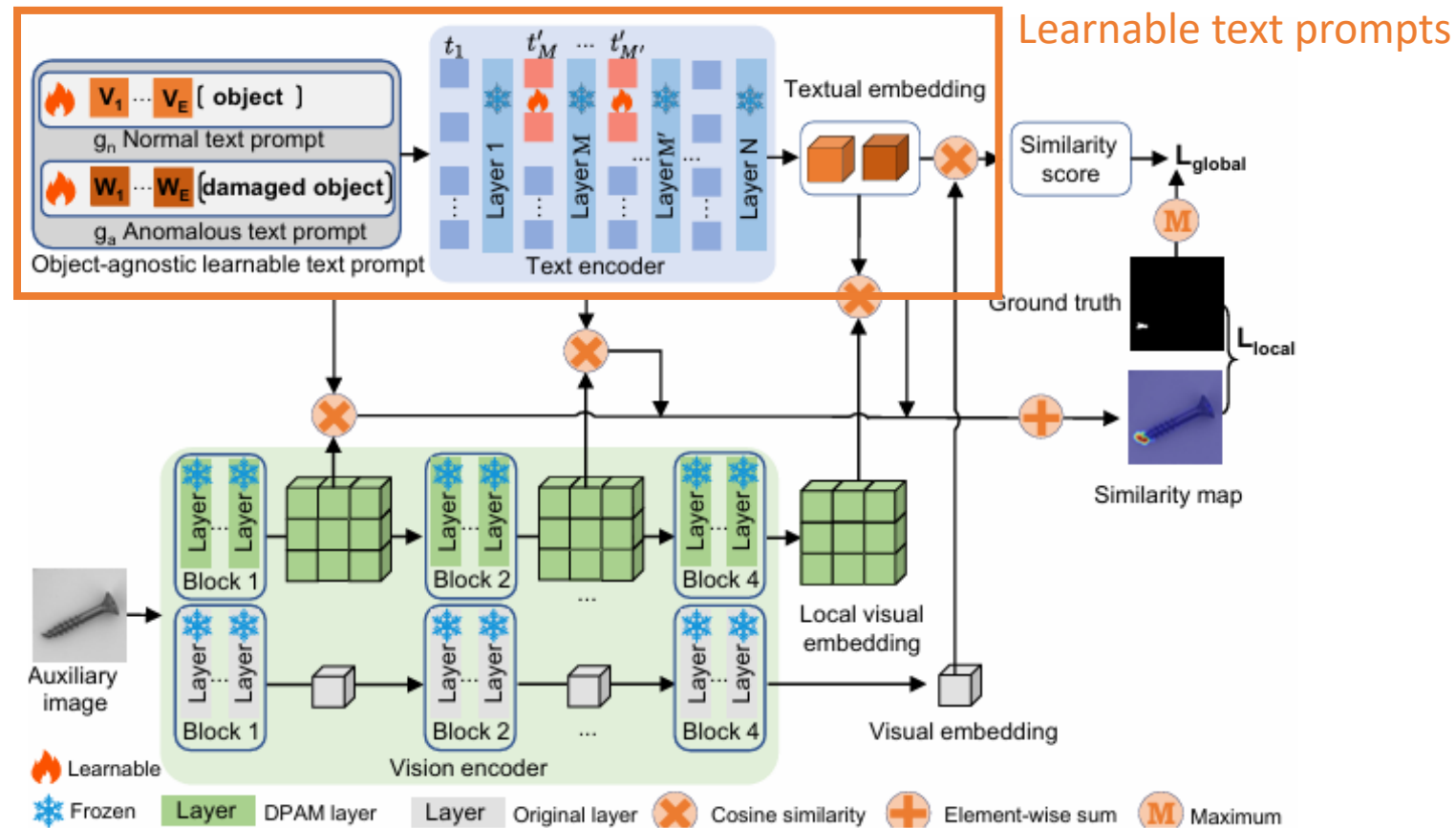
3. Related Work

CLIP-based method designed to address anomaly detection
(Focus on Anomaly Score)



▲ Fig.3 Workflows of WinCLIP/WinCLIP+ (upper/entire pane)

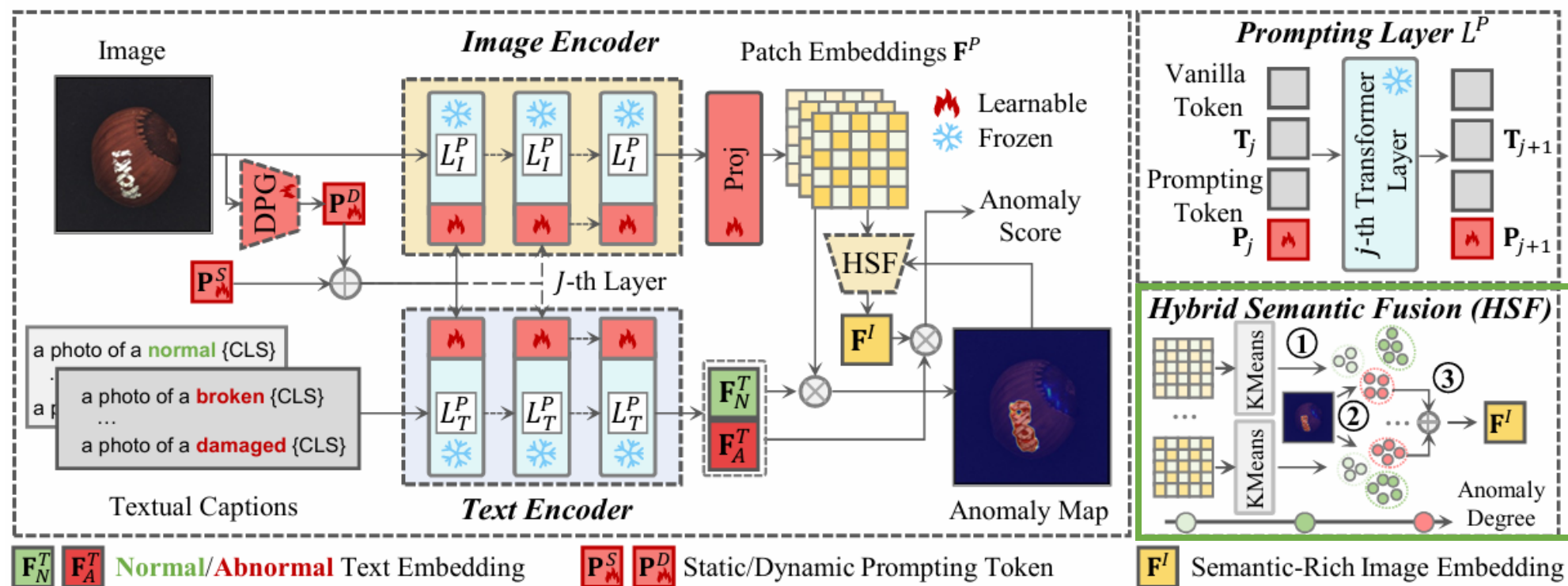
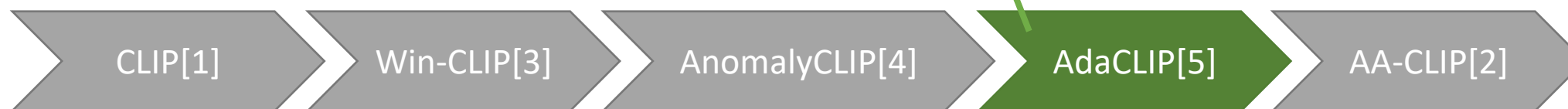
3. Related Work



▲ Fig.4 Workflows of AnomalyCLIP

3. Related Work

Score fusion / Semantic aggregation
(Focus on Anomaly robustness)

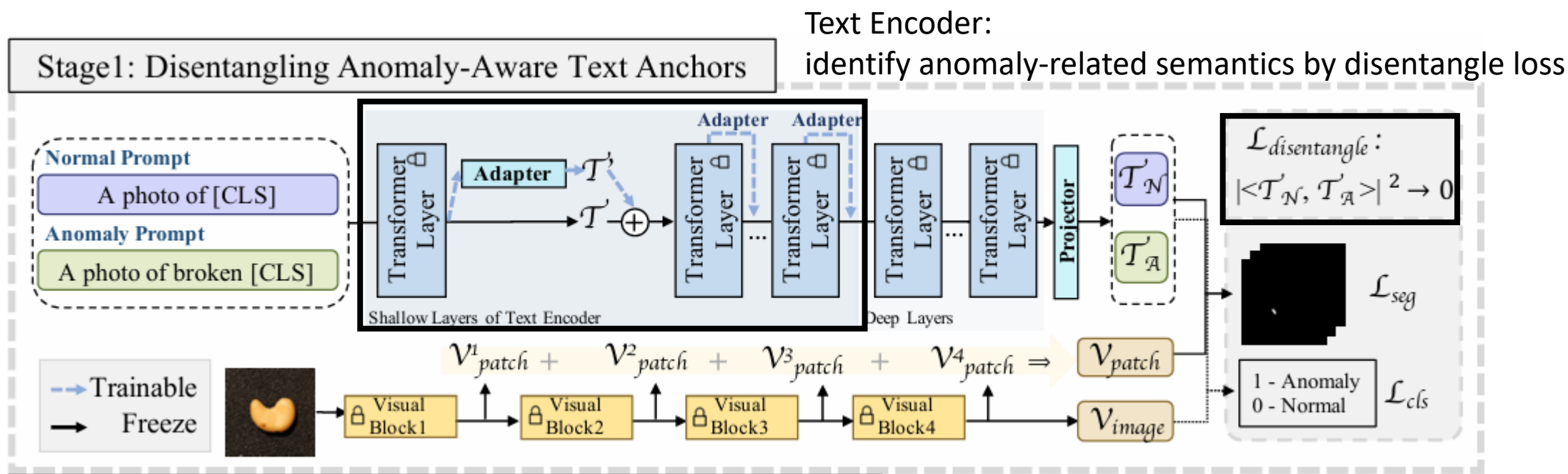
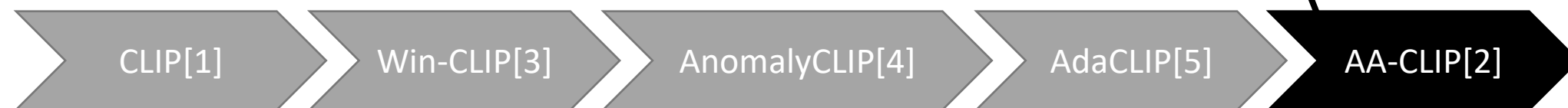


▲ Fig.5 Workflows of AdaCLIP

HSF Module:
Improve image-level AD performance

3. Related Work

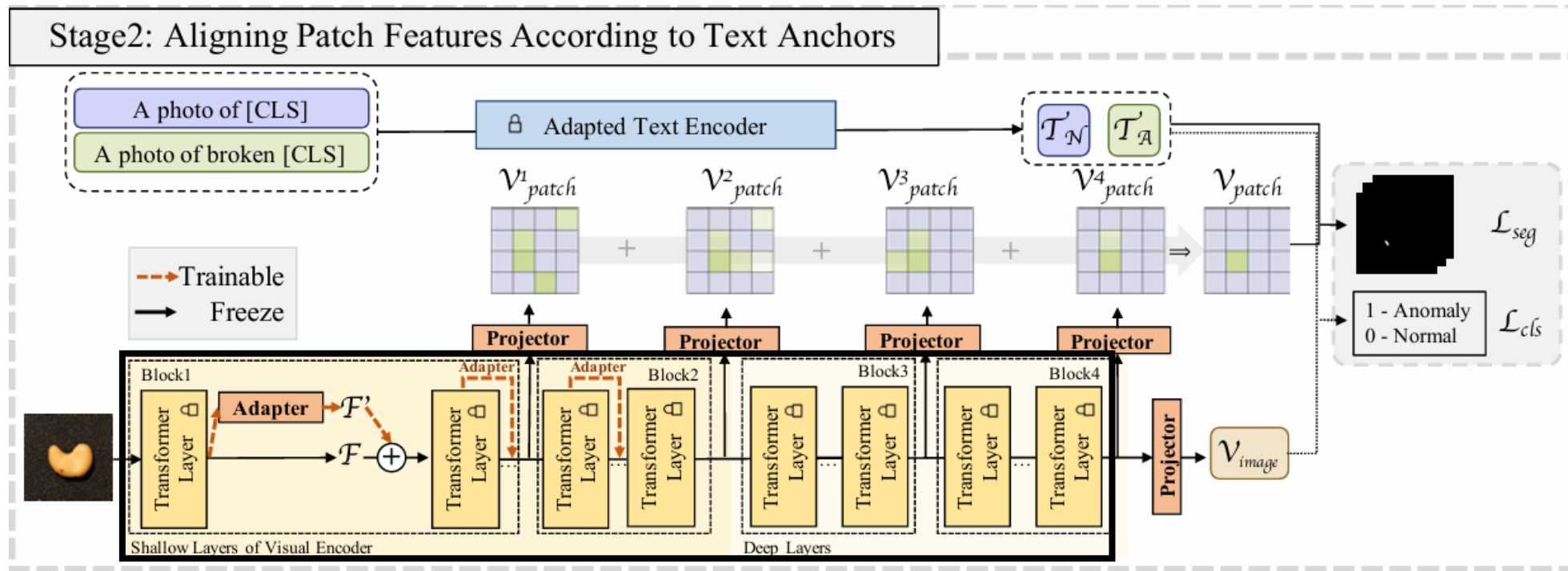
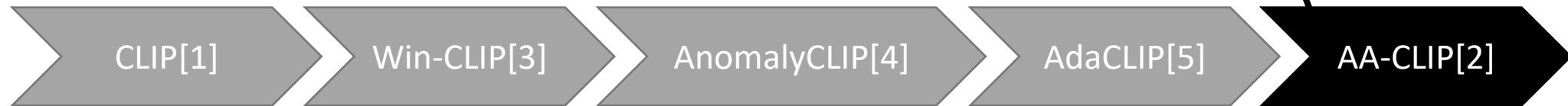
Address both image-level anomaly detection (stage1) and pixel-level anomaly segmentation (stage2)



▲ Fig.6 The Two-Stage Training Pipeline of Anomaly-Aware CLIP

3. Related Work

Address both image-level anomaly detection (stage1) and pixel-level anomaly segmentation (stage2)



Visual(Image) Encoder ▲ Fig.6 The Two-Stage Training Pipeline of Anomaly-Aware CLIP

4. Methodology – Overview

- Based on the framework of AA-CLIP [2], we **replaced 5 main sections** and **used our own methods** to improve the performance

1. Adapter:

Linear Adapter → Frequency-Aware High-Pass Adapter

2. Feature Integration:

Fixed adapt weights for text/visual encoder → Learnable gating

3. Four patches:

Simply summation → learnable `MultiScaleFusion`

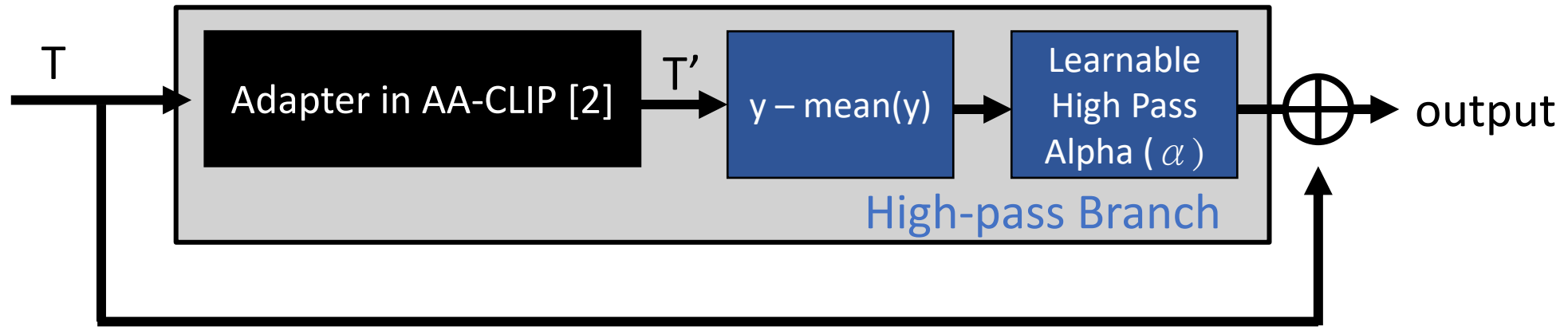
4. Training Strategy (Stage 2 Input):

Standard Image Augmentation → High Freq Augmentation for anomaly areas

5. Inference Strategy (Output):

4. Methodology – Improvement

1. Frequency-Aware High-Pass Adapter



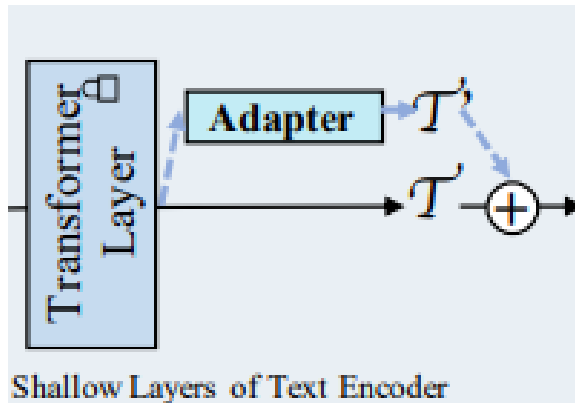
- Take text encoder as an example, this is the same as visual encoder
- Output = $T + T' + \text{High pass noise}$
- **Enhance the model's sensitivity to subtle defects**



4. Methodology – Improvement

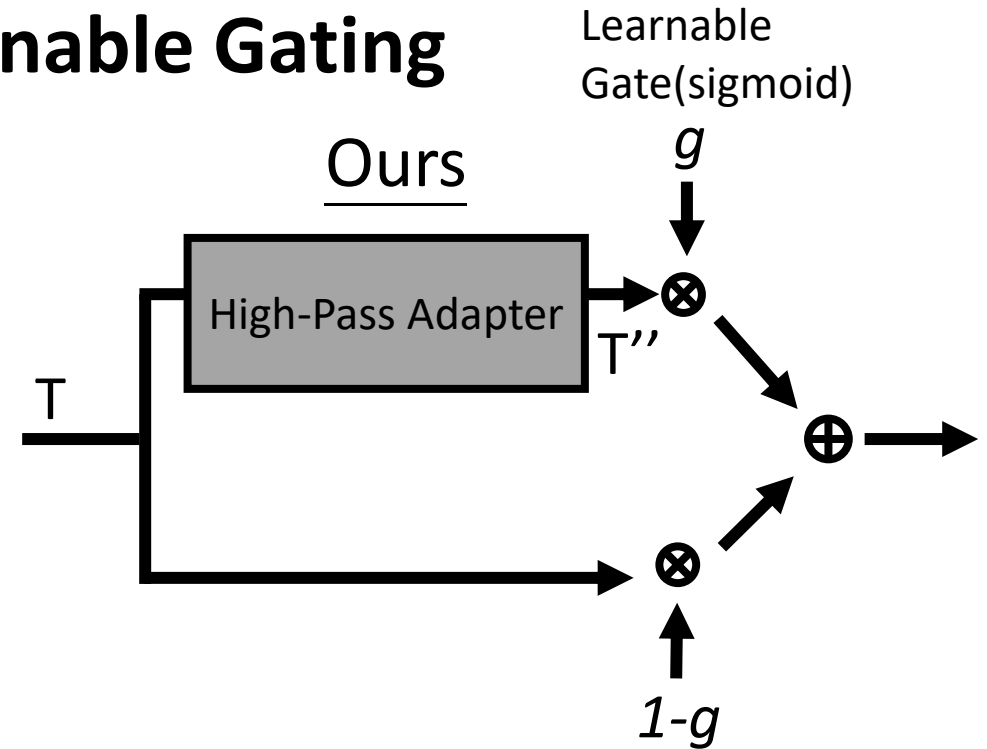
2. Layer-wise Learnable Gating

AA-CLIP [2]



$$Output = T + w(\text{fix weight}) * T'$$

Ours



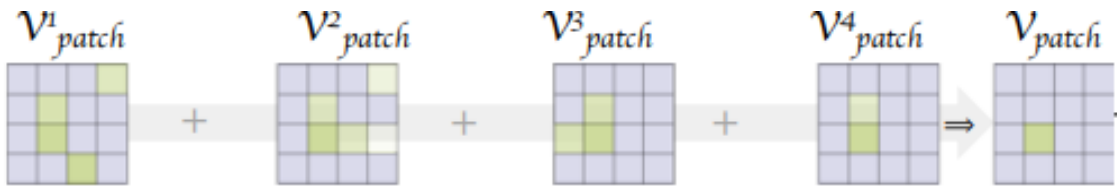
$$Output = g * T'' + (1-g) * T$$

- Take text encoder as an example, this is the same as visual encoder

4. Methodology – Improvement

3. Learnable Multi-Scale Fusion

AA-CLIP [2]



$$V^1 + V^2 + V^3 + V^4 = V_{patch}$$

- Simply summation

Ours

$$V_{patch} = \sum_{i=1}^4 \left(V^i \times \text{softmax}(w_i) \right)$$

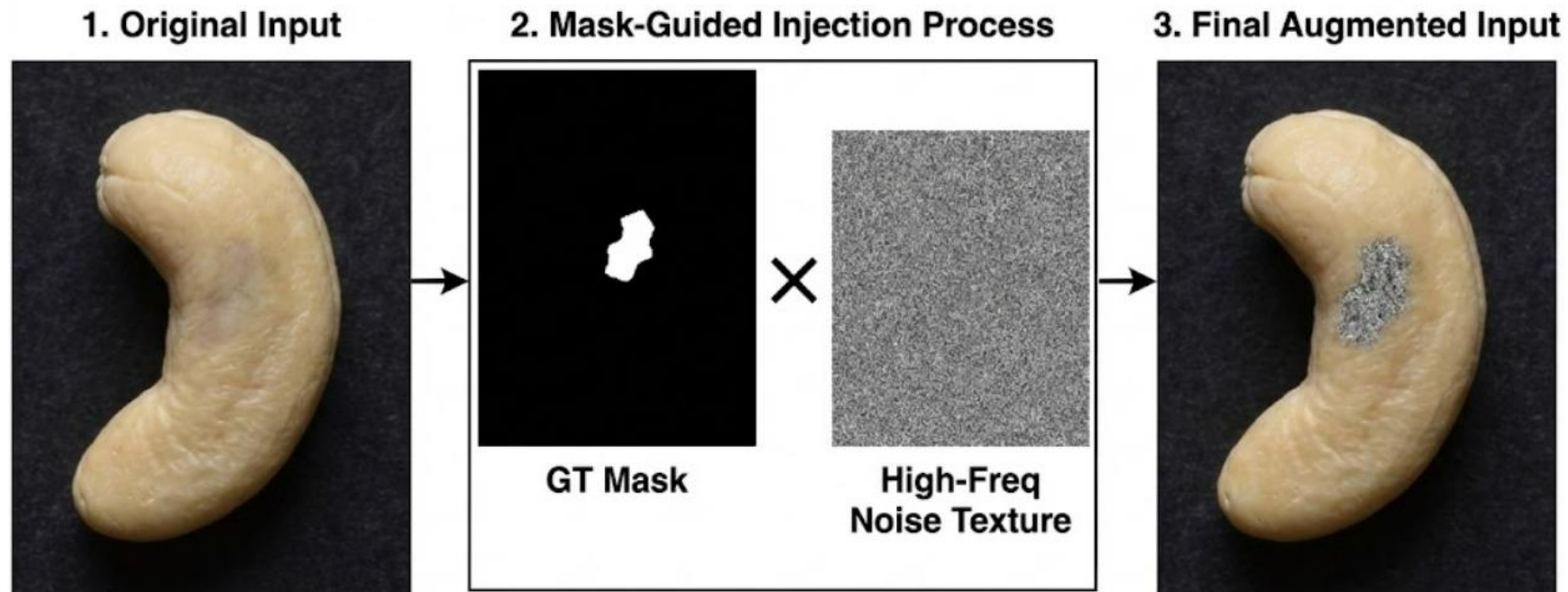
↑
Learnable parameter

- Weighted summation

learnable `MultiScaleFusion`

4. Methodology – Improvement

4. Amonaly High-Freq Augmentation (stage2 input)



- Use the **Ground Truth Mask** to inject high-frequency noise **exclusively** into the anomaly regions.

4. Methodology – Improvement

5. Inference Strategy

AA-CLIP [2]

- Compute Similarity: `matmul()`
Time consuming!
- Anomaly Scoring: $(A+1-N) / 2$
- Max Pooling

Ours

- Compute Similarity: `normalize + einsum()` → compute faster!
- Anomaly Scoring: $A-N$
- Top-k Pooling
→ focus on **regional anomalies** which enhance its **robustness against noise**



5. Experiment Setup

- Training Dataset
 1. ViSA (Visual Anomaly)
 2. MVTec-AD (Industrial Anomaly)
- Target Dataset (1-4: Industrial ; 5-8: Medical)
 1. MVTec-AD / VisA (Cross-validation)
 2. MPDD (Painted Defects)
 3. BTAD (Tech Anomaly)
 4. Brain MRI
 5. Liver
 6. Retina
 7. Colonoscopy (4 sub-datasets: clinicDB, colonDB, Kvasir, CVC-300)
- Metrics: Pixel-AUROC, Image-AUROC (same as AA-CLIP[2])

6. Experimental Results



Falcon-CLIP vs AA-CLIP

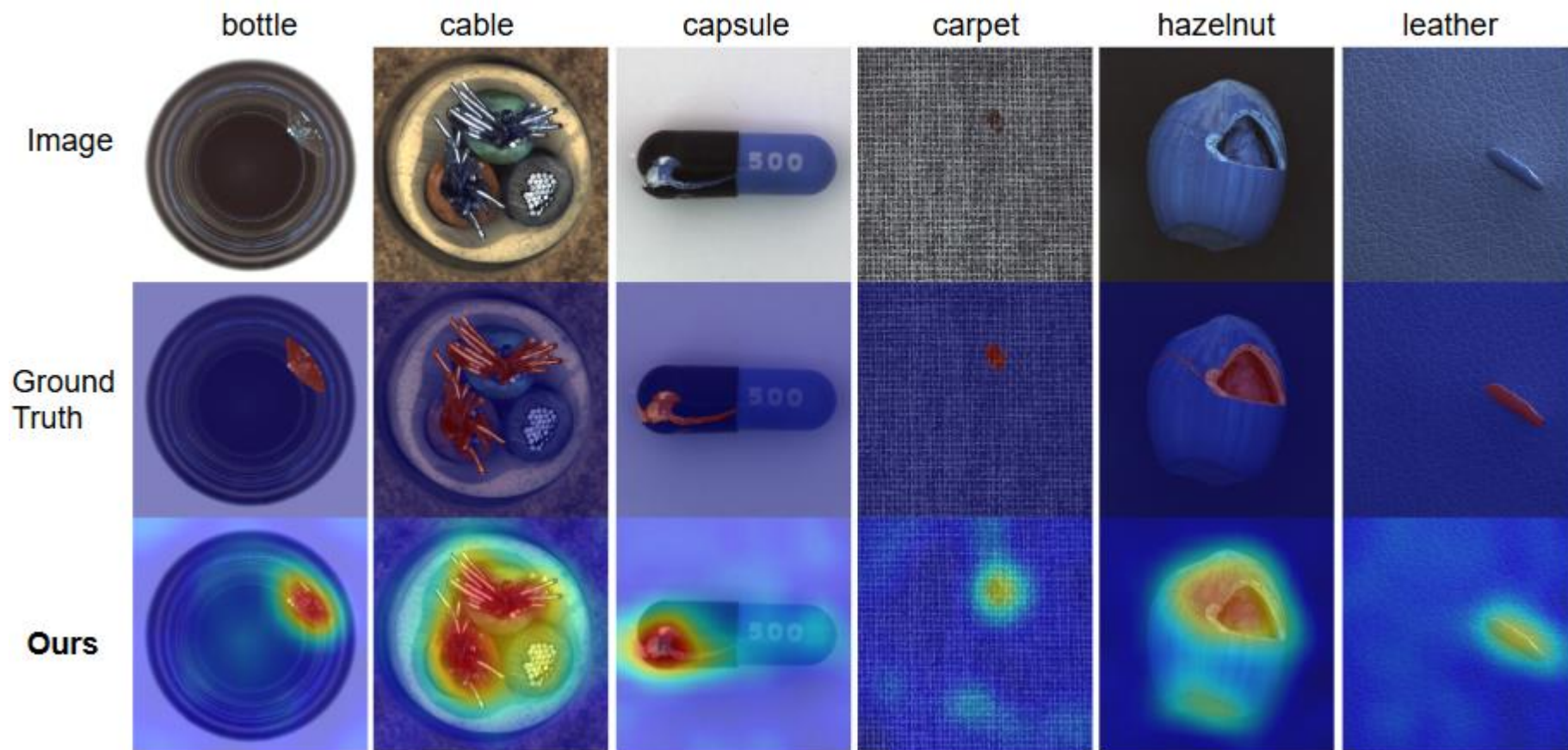
		Ours		Paper	
Sourec DS	Target DS	Pixel AUROC	Image AUROC	Pixel AUROC	Image AUROC
ViSA	MVTec	92.25	91.79	91.9	90.5
ViSA	MPDD	97.05	88.71	96.1	75.1
ViSA	BTAD	96.48	91.59	97	94.8
ViSA	Brain	96.65	78.75	95.5	80.2
ViSA	Liver	96.67	59.86	97.8	69.7
ViSA	Retina	97.05	88.71	95.5	82.7
ViSA	Colon_clinicDB	89.93	NA	89.9	NA
ViSA	Colon_colonDB	83.41	NA	84	NA
ViSA	Colon_Kvasir	87.1	NA	87.2	NA
ViSA	Colon_cvc300	95.47	NA	96.4	NA
MVTec	ViSA	95.53	82.1	95.5	84.6
	Average	93.41727273	83.07285714	93.34545455	82.51428571

6. Experimental Results



Falcon-CLIP vs AA-CLIP

MVTEC: Success case

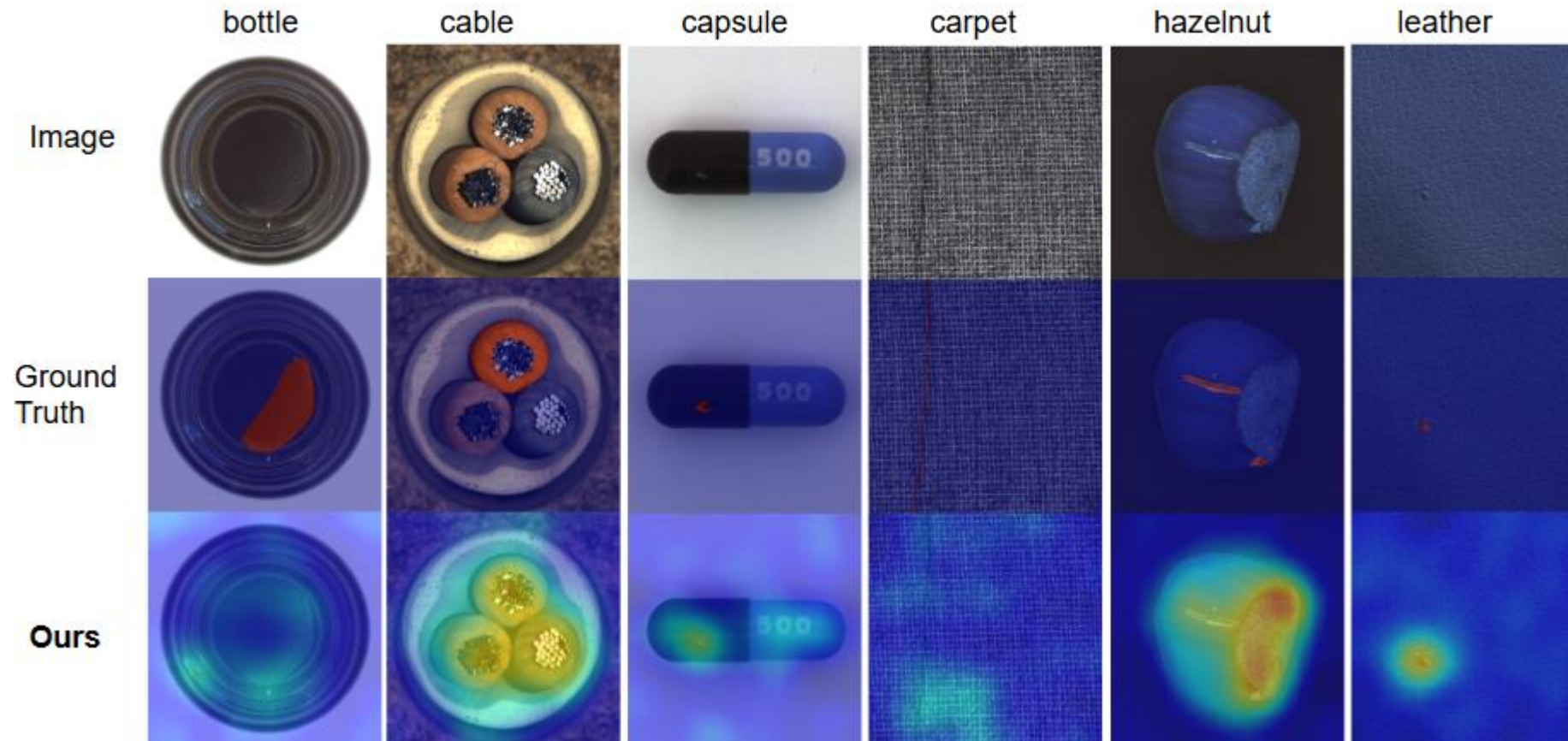


6. Experimental Results



Falcon-CLIP vs AA-CLIP

MVTEC: Failure case

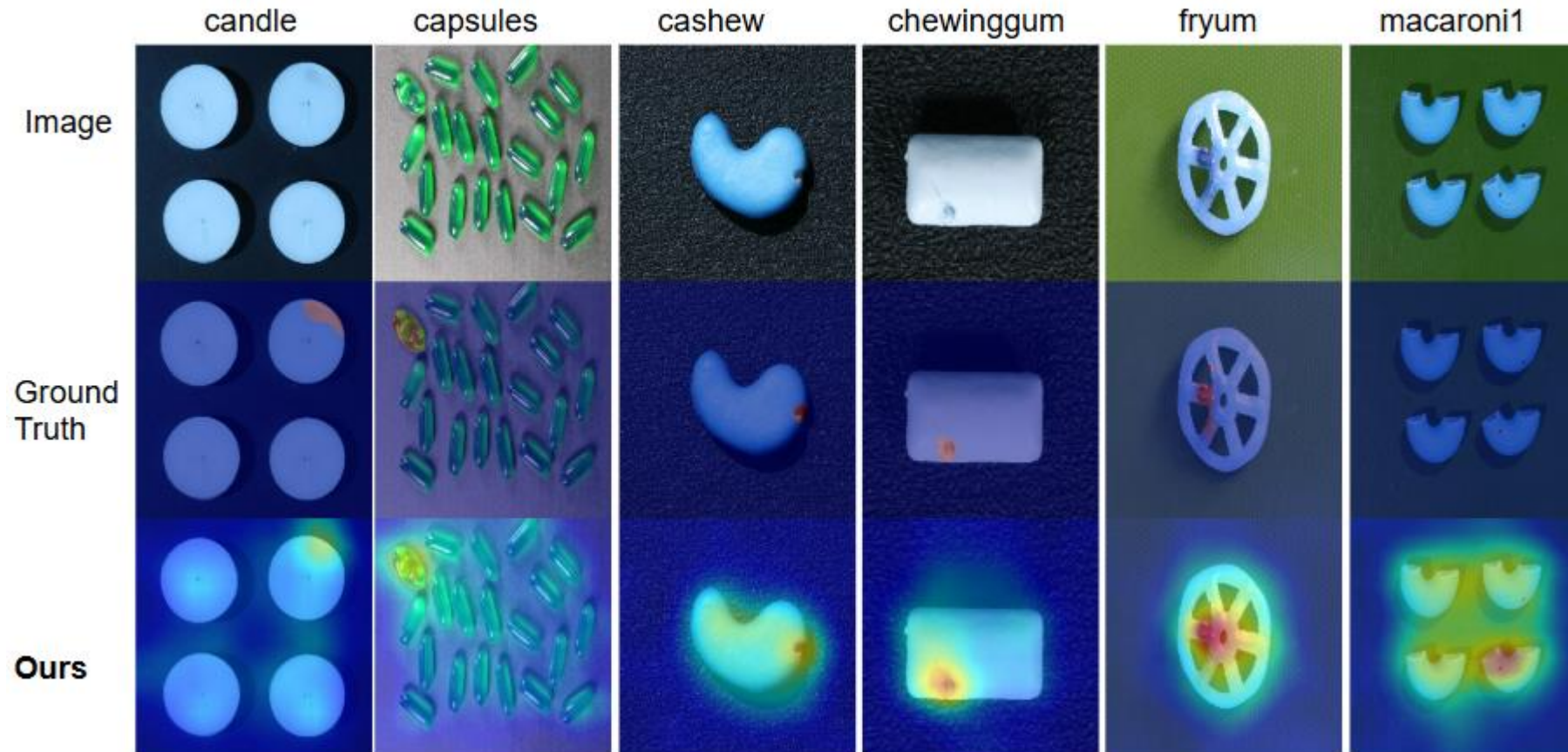


6. Experimental Results



Falcon-CLIP vs AA-CLIP

ViSA : Success case

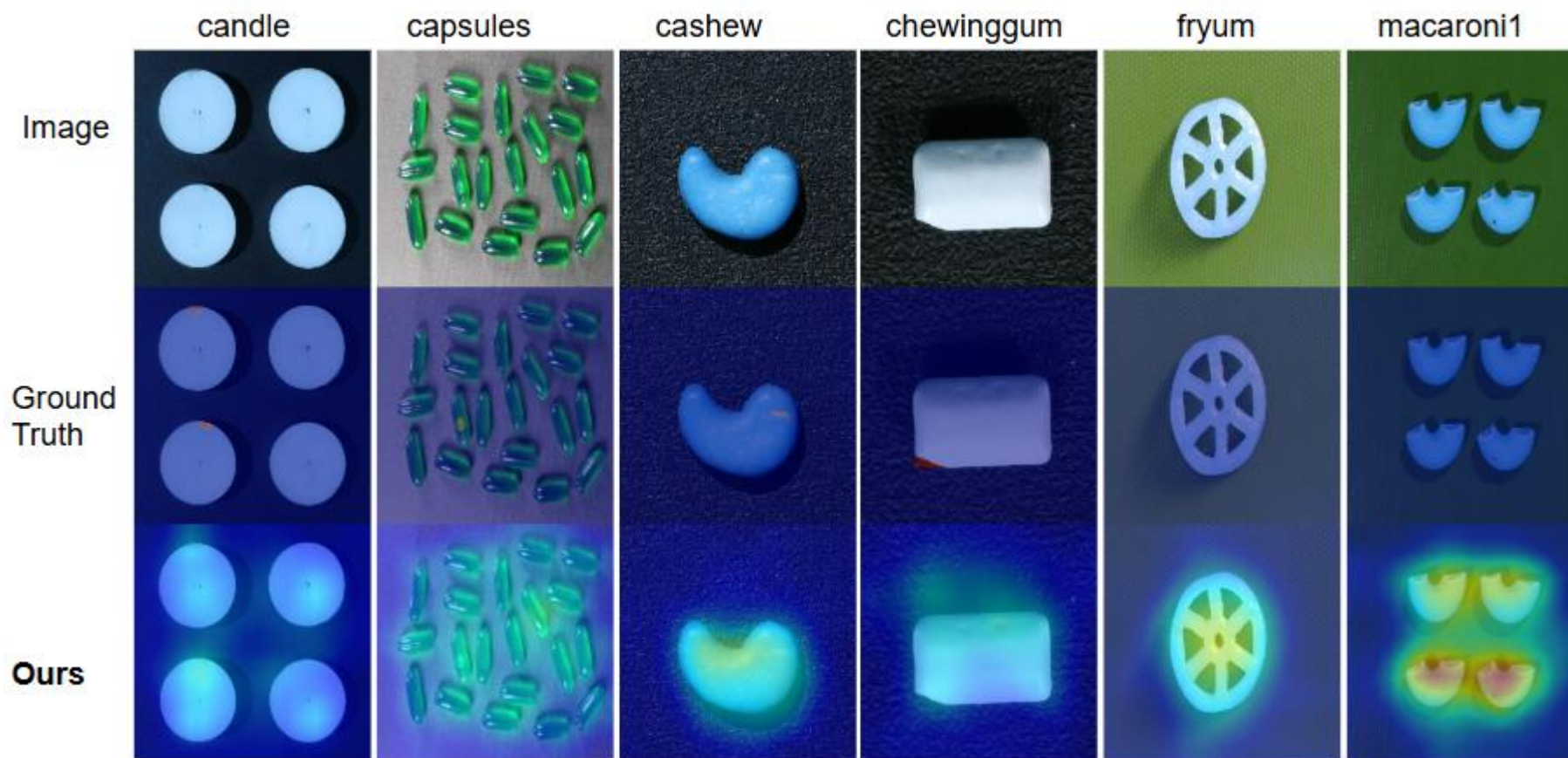


6. Experimental Results



Falcon-CLIP vs AA-CLIP

ViSA : Failure case



7. Conclusion

Achieved remarkable **Pixel-AUROC (Avg 93.41%) / Image-AUROC (Avg 83.07%)** , consistently outperforming baseline (93.34% and 82.51%)

→ Successfully taught a generic CLIP model to '**see**' **high-frequency industrial defects** and **generalize that vision to medical imaging** with pinpoint accuracy.

→ While our model excels at texture anomalies, it sometimes struggles with **extremely subtle, low-contrast point defects** (e.g., tiny dots on *Candle* or *Chewing gum*)

8. Reference

- [1] Alec Radford, Jong Wook Kim, Chris Hallacy, Aditya Ramesh, Gabriel Goh, Sandhini Agarwal, Girish Sastry, Amanda Askell, Pamela Mishkin, Jack Clark, et al. 2021. Learning transferable visual models from natural language supervision. In International conference on machine learning. PMLR, 8748–8763.
- [2] Wenxin Ma, Xu Zhang, Qingsong Yao, Fenghe Tang, Chenxu Wu, Yingtai Li, Rui Yan, Zihang Jiang, and S. Kevin Zhou. 2025. AA-CLIP: Enhancing Zero-shot Anomaly Detection via Anomaly-Aware CLIP. In Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition.
- [3] Jongheon Jeong, Yang Zou, Taewan Kim, Dongqing Zhang, Avinash Ravichandran, and Onkar Dabeer. 2023. Winclip: Zero-/few-shot anomaly classification and segmentation. In Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition. 19606–19616.
- [4] Qihang Zhou, Guansong Pang, Yu Tian, Shibo He, and Jiming Chen. 2024. AnomalyCLIP: Object-agnostic Prompt Learning for Zero-shot Anomaly Detection. In The Twelfth International Conference on Learning Representations.
- [5] Yunkang Cao, Jiangning Zhang, Luca Frittoli, Yuqi Cheng, Weiming Shen, and Giacomo Boracchi. 2024. Adaclip: Adapting clip with hybrid learnable prompts for zero-shot anomaly detection. In European Conference on Computer Vision. Springer, 55–72.

Thank you for listening !